

Bernoulli Trials

In our introduction activity, we went over the requirements for labeling an event a “Bernoulli Trial”:

- 1.
- 2.
- 3.

If we take a Bernoulli Trial (like our true/false quiz) and then we are interested in the number of successes in a specific number of trials, we create a **Binomial Probability Distribution**.

Binomial Probability Distribution

There are four requirements for a setting to follow the binomial probability distribution. As long as these four things are true, we can use the binomial probability model for calculating the probabilities.

- Two Possible Outcomes
 - Each trial falls into one of two categories – we call them “success” or “failure.”
 - Success does not necessarily mean a positive outcome, but instead, the outcome we are looking for.
- Fixed Probability
 - The probability of success (called p) is the same for each trial.
- Independent Trials
 - Each trial is independent of the next.
 - Reasonable Assumption for coins, cards with replacement, spinners, rolling a die, etc.
 - What about sampling without replacement, when it can’t be avoided? This is where the “10% condition” comes in.
 - The 10% condition says that if you are sampling from a large enough population, you can proceed with sampling without replacement as though the trials were independent
 - Ex. Sampling from a deck of cards without replacement would violate this but sampling people from a large town would be okay.
- Fixed Trials
 - We are counting the number of successes from a set number of trials (called n).

If all of the four points above are satisfied, you can compute the probability you get a certain number of successes in a specified number of trials:

n = number of independent trials p = probability of success x = number of successes	$P(X = x) =$ where	Mean: The expected number of successes Standard Deviation: The typical deviation from the expected number of successes
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Why this formula?

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

Examples

Directions: For the following situations, explain how each of the settings follows the binomial probability distribution model.

1) You toss a coin 10 times and count the number of tails you get.	$X =$ Two Possible Outcomes: Fixed Probability: Independent Trials: Fixed Trials:	Find and interpret $P(X = 4)$
2) A company reports that 75% of emails it receives is spam. You have 25 emails in your inbox.	$X =$ Two Possible Outcomes: Fixed Probability: Independent Trials: Fixed Trials:	Find and interpret $P(X = 20)$
3) Type A blood is found in 43% of the population. You have a group of 40 students.	$X =$ Two Possible Outcomes: Fixed Probability: Independent Trials: Fixed Trials:	Find and interpret $P(X = 15)$

Binomials on the Calculator

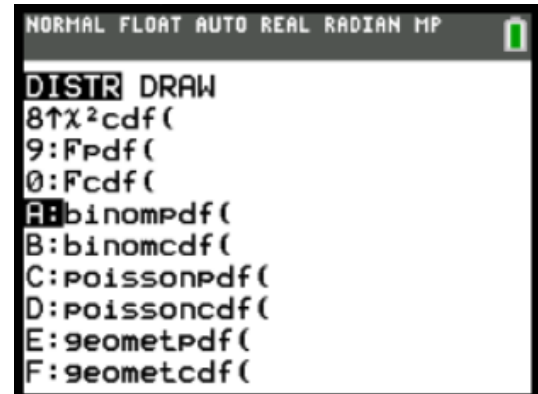
On your calculator, you will see a lot of distributions with “pdf” and “cdf” after it and THERE IS A DIFFERENCE!!!

_____ = probability distribution function and is used with the distribution when you are trying to calculate is $P(X = x)$

_____ can be computed using `binompdf(n, p, x)`

_____ = cumulative distribution function and is used with the distribution when you are trying to calculate the probability of an interval of values

_____ can be computed using `binomcdf(n, p, x)`



4) Type A blood is found in 43% of the population. You have a group of 40 students. What is the probability that you have exactly 20 people in your group with Type A blood?

5) Type A blood is found in 43% of the population. You have a group of 40 students. What is the probability that 20 people or less in our group have Type A blood?

- A binomial random variable is _____
- Inequalities matter when finding probabilities
- Your calculator can only do binomcdf with _____, so all inequalities (if you use your calculator) must be converted to that inequality.
- We use the idea of a _____ to find probabilities that do not involve “=” or “≤”

Examples

Type A blood is found in 43% of the population. You have a group of 40 students.

6) What is the probability that 14 or more students have Type A blood?

7) What is the probability that less than 10 students have Type A blood?

8) What is the probability that more than 30 students have Type A blood?